

Lucid dreaming occurs when a person becomes consciously aware of a dream and uses executive decision-making skills to critically analyze the dream environment. Lucid dreamers are thus able to control certain aspects of their dreams, an ability that may occur intentionally, through practice, or unintentionally, usually after having noticed something bizarre or unrealistic in the dream. A lucid dreamer might choose to experience the sensation of flying, explore a custom-designed landscape, carry out a predetermined task, or practice a skill. Estimates suggest that about half the population has had a lucid dream, and about 10%-20% of people are considered frequent lucid dreamers experiencing such dreams regularly (at least once a month).

In one study, researchers asked participants to perform a difficult visuomotor task that involved throwing an object through a small moving target. Participants were organized into three groups, one group of frequent lucid dreamers and two groups of nonlucid dreamers. All three groups attempted the visuomotor task at midday in the research lab, before (pretest) and after (posttest) a single night's sleep at home. Lucid dreamers were asked to only practice the visuomotor task overnight during their lucid dream ("lucid practice" group). One group of nonlucid dreamers practiced the task in the lab, right after the pretest ("waking practice" group), and another group of nonlucid dreamers performed the pretest and posttest without practicing between the tests ("no practice" group). The average improvement of hits (the number of times each participant successfully threw the object through the target) from pretest to posttest for each group is shown in Figure 1.

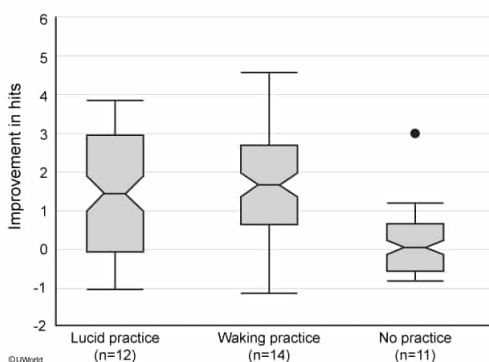
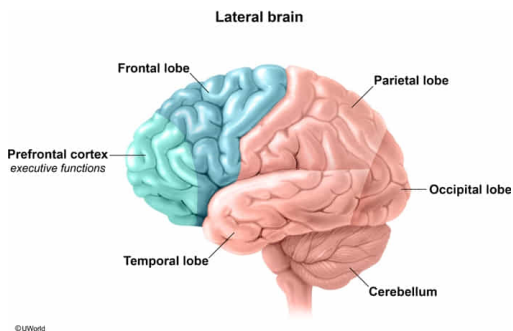


Figure 1 Median improvement in total hits for each group (Note: The notches represent the 95% confidence interval for the median.)

The researchers proposed that a follow-up study be devised and conducted to control for any potential variability in each participant's sleep at home. The researchers suggested using the same study procedure except participants would sleep overnight for five consecutive nights in a sleep lab, where the timing and duration of sleep would be standardized. In order to eliminate any potential "first-night effects" (an initial reduction in sleep quality when sleeping in an unfamiliar environment), researchers proposed that pretest and posttest measures each be taken before and after the third, fourth, and fifth night's sleep.

If a neuroimaging study compared normal dreaming with lucid dreaming, in what brain region would more activity be expected during lucid dreaming?

- ☐ A. Cerebellum [5%]
- ✓ ☒ B. Prefrontal cortex [84%]
- ☐ C. Amygdala [2%]
- ☐ D. Primary visual cortex [7%]

Explanation:

The **prefrontal cortex** is associated with *executive functions*, such as **critical thinking**, **problem solving**, **planning**, **impulse control**, and **executive decision-making**. Because lucid dreaming involves critical analysis and executive decision making, a neuroimaging study would be expected to demonstrate greater activation of the prefrontal cortex during lucid dreaming as compared with normal dreaming.

The prefrontal cortex is the anterior portion of the **frontal lobe**, which also contains the primary motor cortex (responsible for the initiation of voluntary movement) and Broca area (involved in speech production).

(Choice A) The cerebellum is positioned underneath the occipital lobe at the back of the skull and is associated with motor coordination and motor learning. This region would not necessarily be more active during lucid dreaming compared with normal dreaming.

(Choice C) The amygdala is part of the limbic system located on either side of the thalamus deep in the cortex; it is associated with processing emotion, particularly fear and anger. Although this region may be active during lucid dreaming, it is probably no more active than during normal dreaming.

(Choice D) Located in the occipital lobe, the primary visual cortex is associated with processing visual input from the retinas and is not expected to be as active during dreaming, when there is no external visual input. However, secondary visual areas of the brain have been associated with the visual imagery that occurs during dreaming.

Educational objective:

The prefrontal cortex is involved in executive decision making and critical analysis. It is located in the frontal lobe, which also contains the primary motor cortex.

Time spent:0 sec

QID:400134

Subject:
Behavioral Sciences

System:
Sensation, Perception, and Consciousness

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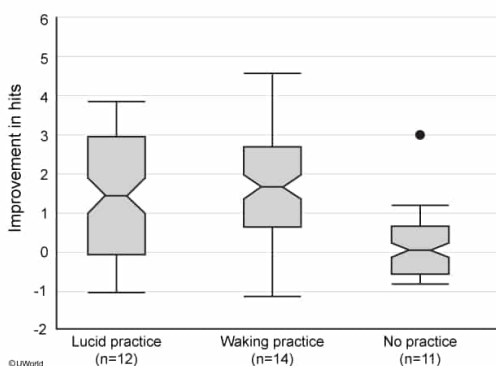


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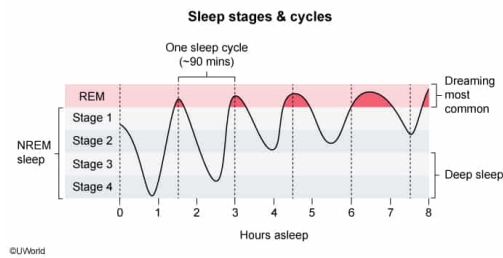
If a lucid dreamer falls asleep at 10:00 PM and wakes up at 6:00 AM, during what period of sleep would this person have the greatest chance of lucid dreaming?

- ☐ A. Between 10:00 PM and 11:30 PM [6%]
- ☐ B. Between 1:00 AM and 3:00 AM [35%]
- ✓ ☒ C. Between 4:00 AM and 6:00 AM [46%]
- ☐ D. Chances of lucid dreaming are roughly equal throughout sleep [10%]

Correct

45% answered correctly

Explanation:



Sleep is broadly divided into **non-rapid eye movement (NREM) sleep** (stages 1-4) and **rapid eye movement (REM)**. One **sleep cycle** includes various proportions of REM and NREM sleep, each lasting about 90 minutes. There are approximately 4-6 sleep cycles in a typical night's sleep.

The amount of time spent in each sleep stage differs over the course of a single night's sleep; a typical adult spends about 75% of sleep time in NREM sleep and 25% in REM sleep. The time spent in REM varies, increasing in duration for each subsequent sleep cycle, with the **last cycle** containing the **most REM** sleep.

Studies suggest that **dreaming** occurs during both REM and NREM sleep but is **most common** during **REM** sleep, particularly during the REM cycles that occur closer to waking. REM dreams tend to be more vivid and memorable and are more likely to be recalled upon waking. Studies suggest lucid dreaming occurs only during REM sleep.

An individual sleeping from 10:00 PM to 6:00 AM would have the greatest chance of lucid dreaming during the last sleep cycle, from 4:00 AM to 6:00 AM, because the greatest proportion of REM sleep occurs just before waking.

(Choice A) The shortest REM period occurs during the first sleep cycle (10:00 PM to 11:30 PM), so there is the *least chance* of lucid dreaming during this time.

(Choice B) The greatest chance of having a lucid dream is during one of the longer REM periods that occur during the last sleep cycle, rather than during REM sleep that occurs in the middle of the night (1:00 AM to 3:00 AM).

(Choice D) Because REM periods increase in duration during the night with each subsequent sleep cycle, the chances of having a lucid dream are not equal throughout the night.

Educational objective:

Dreaming, which occurs during both REM and NREM sleep, is more common, vivid, and easily recalled when it occurs during REM sleep. The duration of REM sleep is shortest earlier in the night and longest during the last sleep cycle, which occurs in the early morning hours.

Time spent: 0 sec
Subject:
Behavioral Sciences

QID: 400135
System:
Sensation, Perception, and Consciousness